Appendix F
Selected Solutions

F.3 Chapter 3 Solutions

3.1

	N-Type	P-Type
Gate=1	closed	open
Gate=0	open	closed

3.3 There can be 16 different two input logic functions.

3.5

A	B	C		OUT
0	0	0		1
0	0	1		0
0	1	0		1
0	1	1		0
1	0	0		1
1	0	1		0
1	1	0		0
1	1	1		0

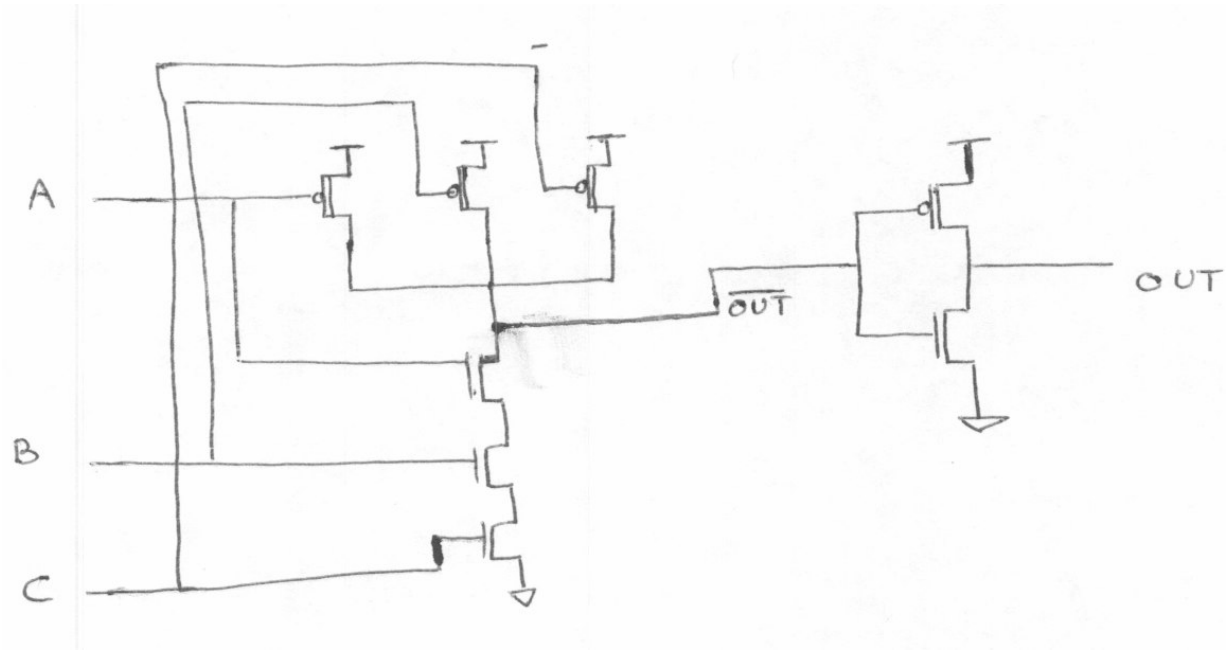
3.7 There is short circuit (path from Power to Ground) when either A = 1 and B = 0 or A = 0 and B = 1.

3.9

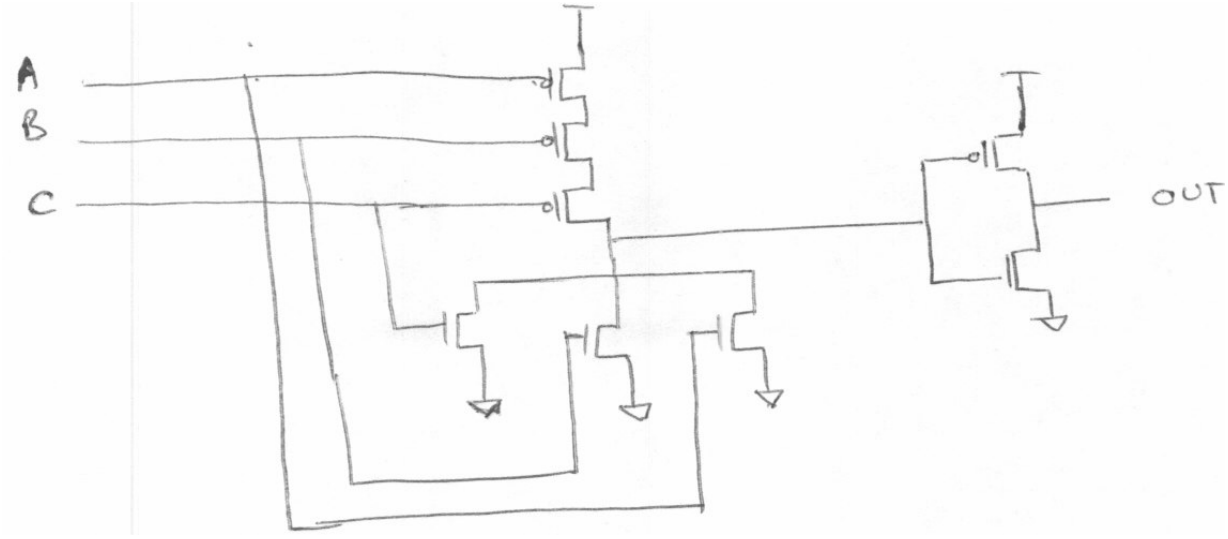
A	B		NOT(NOT(A) OR NOT(B))
0	0		0
0	1		0
1	0		0
1	1		1

AND gate has the same truth table.

3.11 a. Three input And-Gate

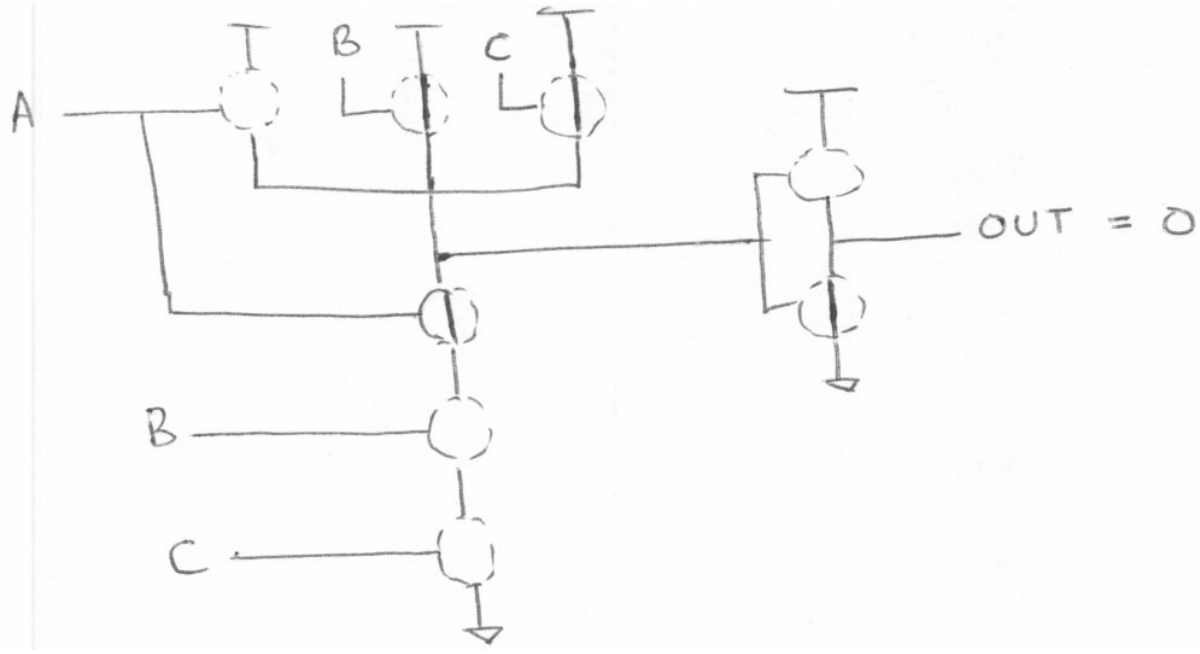


Three input OR-Gate

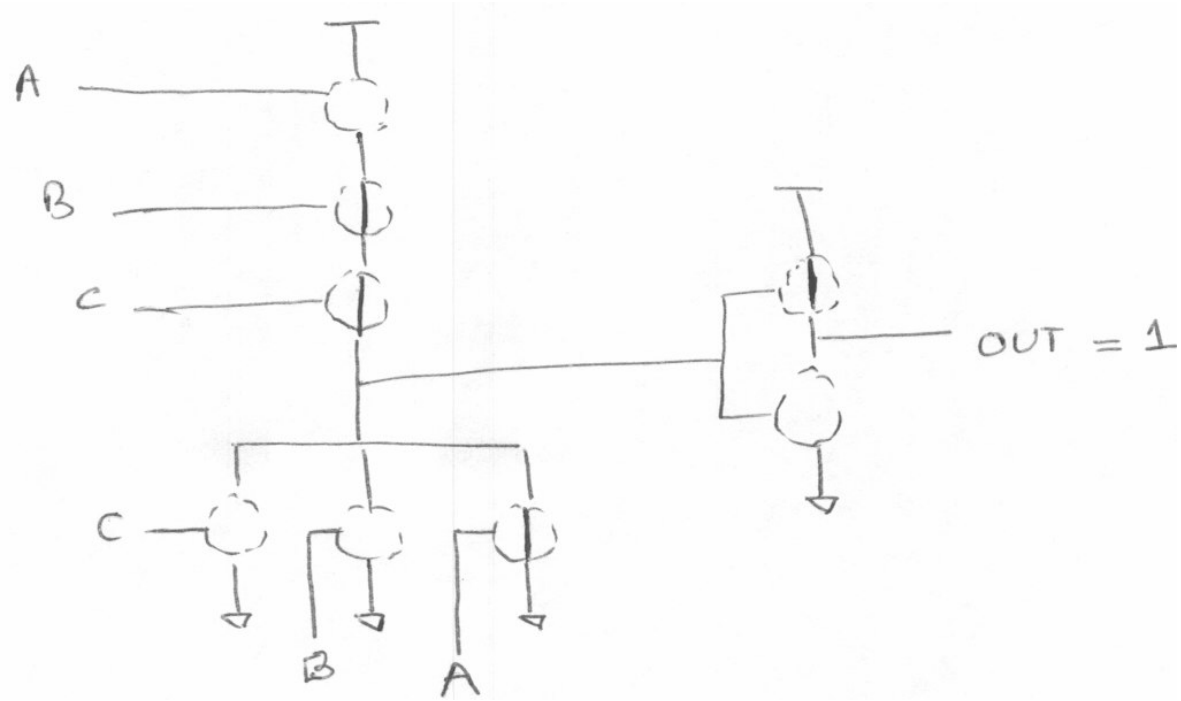


b. (1) A = 1, B = 0, C = 0.

AND Gate

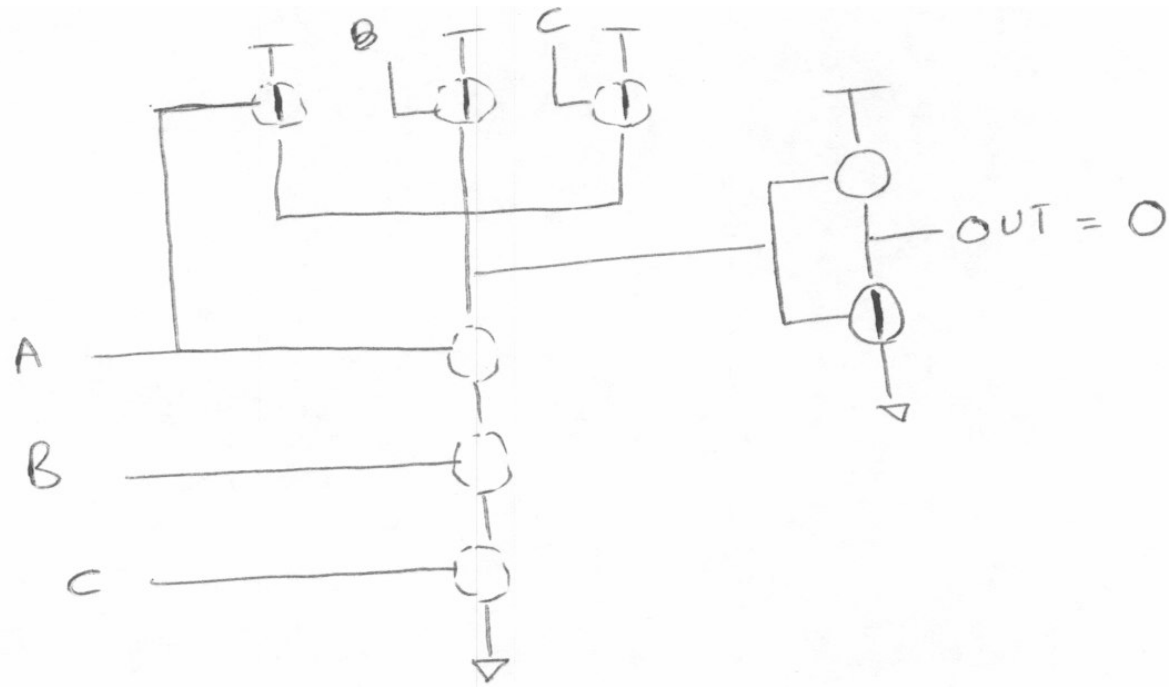


OR Gate

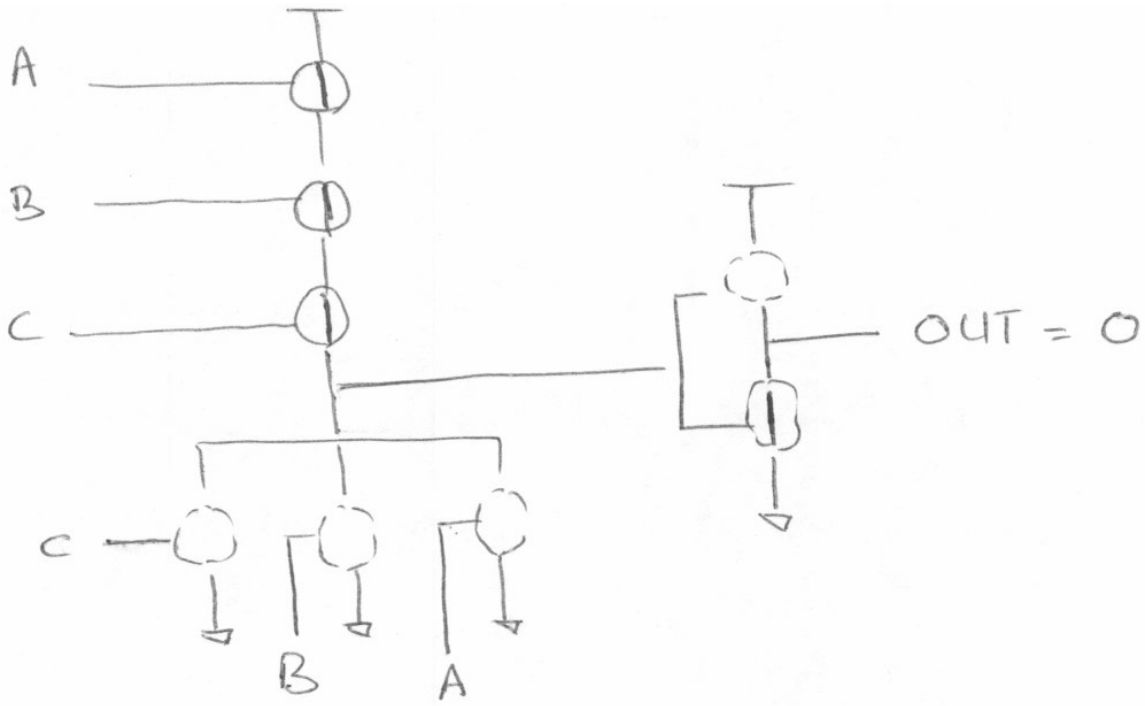


b. (2) A = 0, B = 0, C = 0

AND Gate

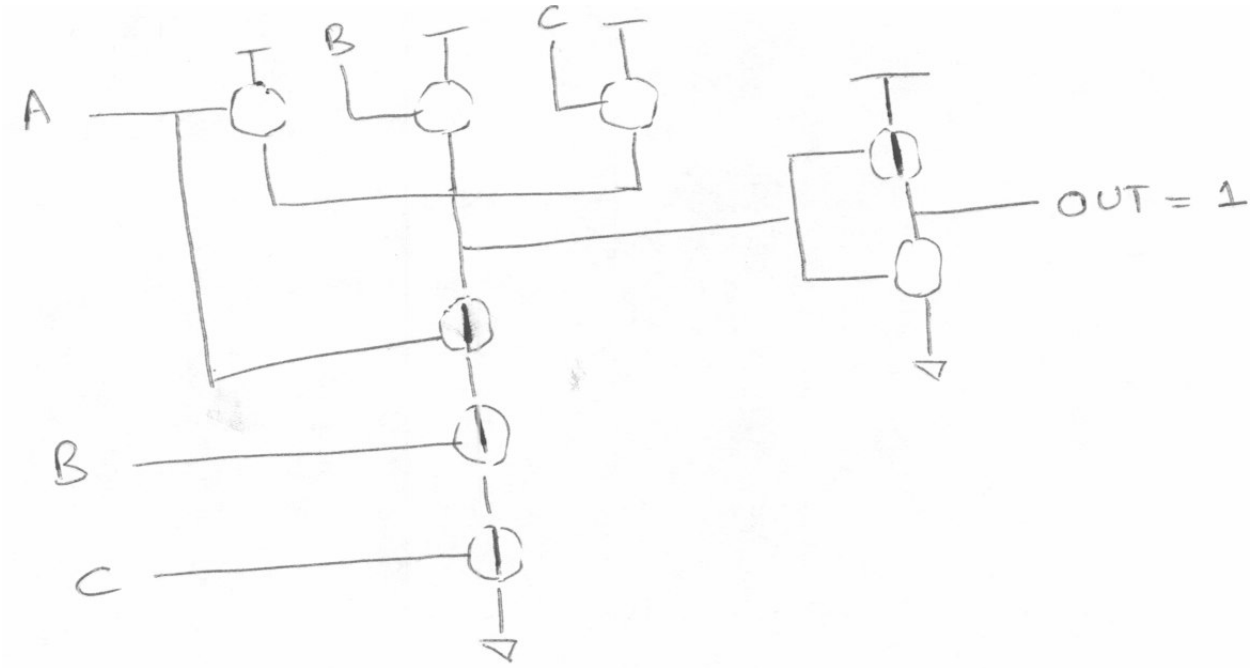


OR Gate

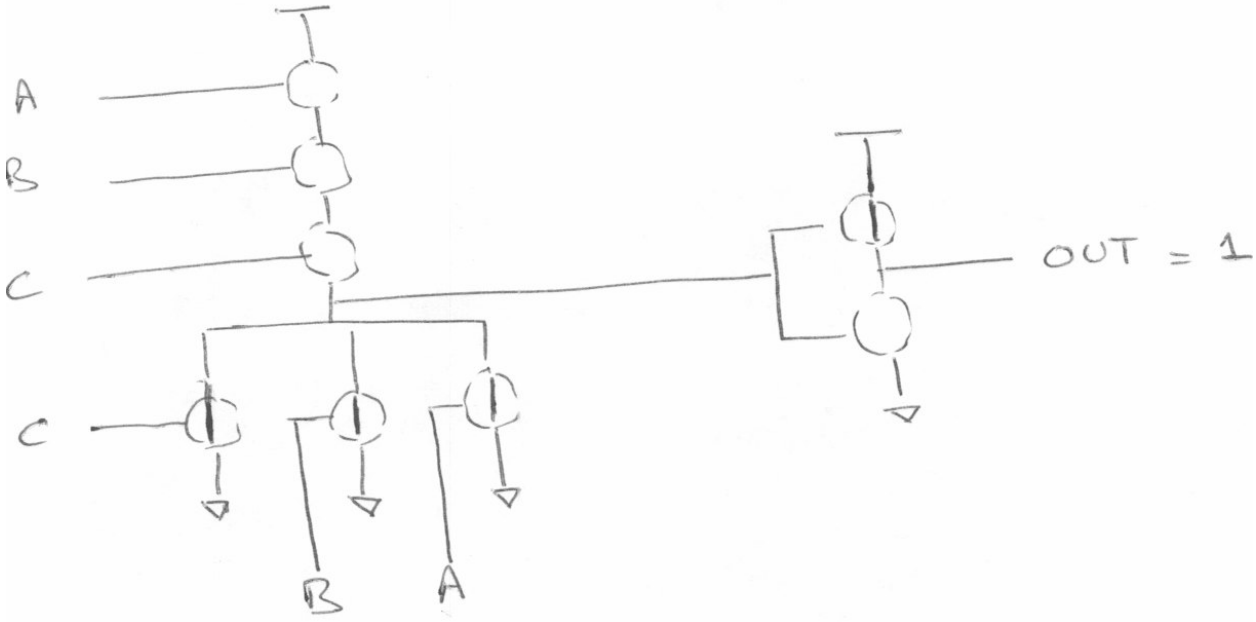


b. (3) A = 1, B = 1, C = 1

AND Gate



OR Gate



3.13 A five input decoder will have 32 output lines.

3.15

C_{in}	1	1	1	0
A	0	1	1	1
B	1	0	1	1
S	0	0	1	0
C_{out}	1	1	1	1

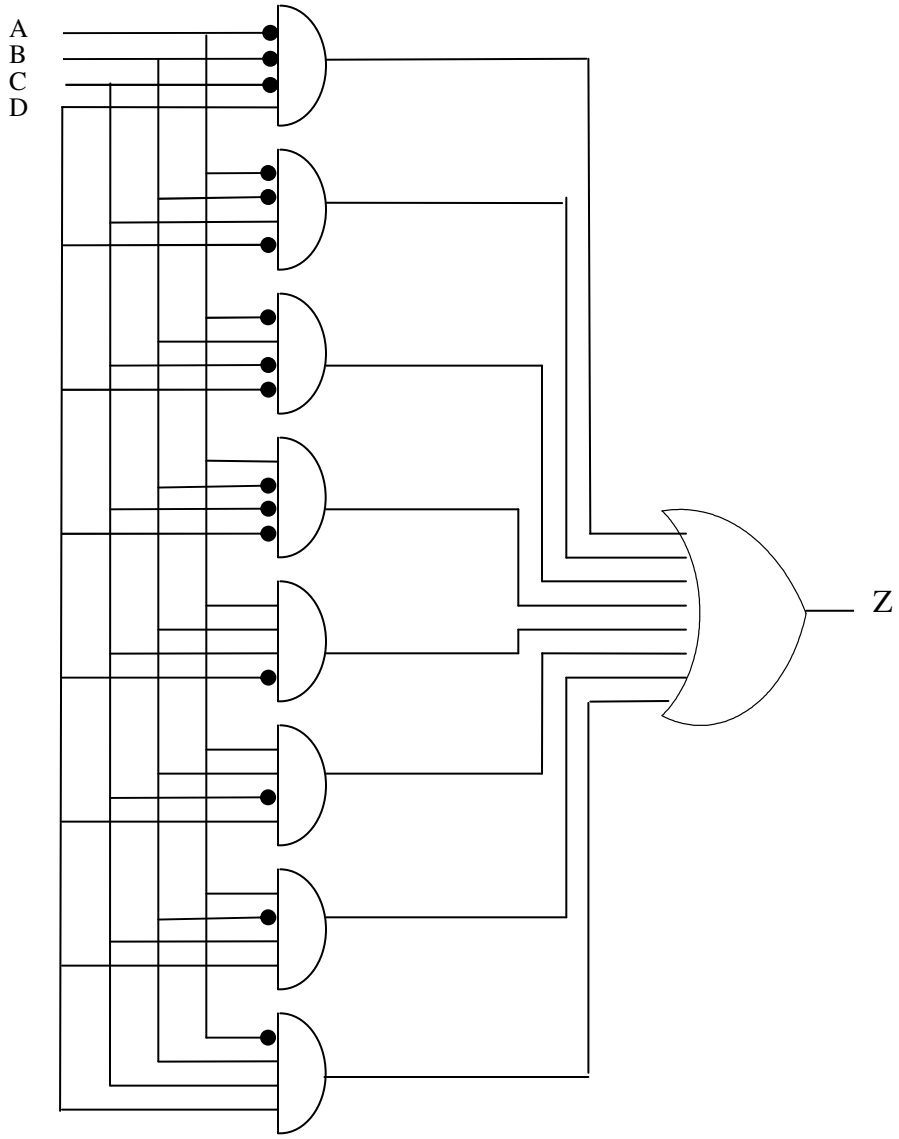
A = 7, B = 11, A + B = 18.
In the above calculation, the result (S) is 2 !! This is because 18 is too large a number to be represented in 4 bits. Hence there is an overflow - Cout[3] = 1.

3.17 (a) The truth table will have 16 rows. Here is the truth table for Z = XOR (A, B, C, D). Any circuit with at least seven input combinations generating 1s at the output will work.

A	B	C	D		Z
0	0	0	0		0
0	0	0	1		1
0	0	1	0		1
0	0	1	1		0
0	1	0	0		1
0	1	0	1		0
0	1	1	0		0
0	1	1	1		1
1	0	0	0		1
1	0	0	1		0
1	0	1	0		0
1	0	1	1		1
1	1	0	0		0
1	1	0	1		1
1	1	1	0		1
1	1	1	1		0

Z = XOR (A,B,C,D)

(b)



3.19 Figure 3.36 is a simple combinational circuit. The output value depends ONLY on the input values as they currently exist. Figure 3.37 is an R-S Latch. This is an example of a logic circuit that can store information. That is, if A, B are both 1, the value of D depends on which of the two (A or B) was 0 most recently.

3.21 $2 * 2^{14} = 2^{15} = 32768$ nibbles

3.23

A	B	C		Z
0	0	0		0
0	0	1		0
0	1	0		0
0	1	1		0
1	0	0		0
1	0	1		0
1	1	0		0
1	1	1		0

3.25 (a) 3 gate delays

3.25 (b) 3 gate delays

3.25 (c) 3*4 = 12 gate delays

3.25 (d) 3*32 = 96 gate delays

3.27(a) When S=0, Z = A

3.27(b) When S=1, Z retains its previous value.

3.27(c) Yes; the circuit is a storage element.

3.29 No. The original value cannot be recovered once a new value is written into a register.

3.31. 8 * (2^3) = 64 bytes

3.33.(a) To read the 4th memory location, A[1,0] = 11, WE = 0

3.33.(b) A total of 6 address lines are required for a memory with 60 locations. The addressability of the memory will remain unchanged.

3.33.(c) A program counter of width 6 can address 2^6 = 64 locations. So without changing the width of the program counter, 64-60 = 4 more locations can be added to the memory.

3.35 Total bits of storage = 2^22 * 3 = 12582912

3.37 There are a total of four possible states in this lock. Any other state can be expressed as one of states A, B, C or D. For example, the state performed one correct followed by one incorrect operation is nothing but state A as the incorrect operation reset the lock.

3.39 No. An arc is needed between the two states.

(a) Game in Progress:

Texas *	Oklahoma
Fouls:4	Fouls: 4
73	68
First Half	
7:38	
Shot Clock : 14	

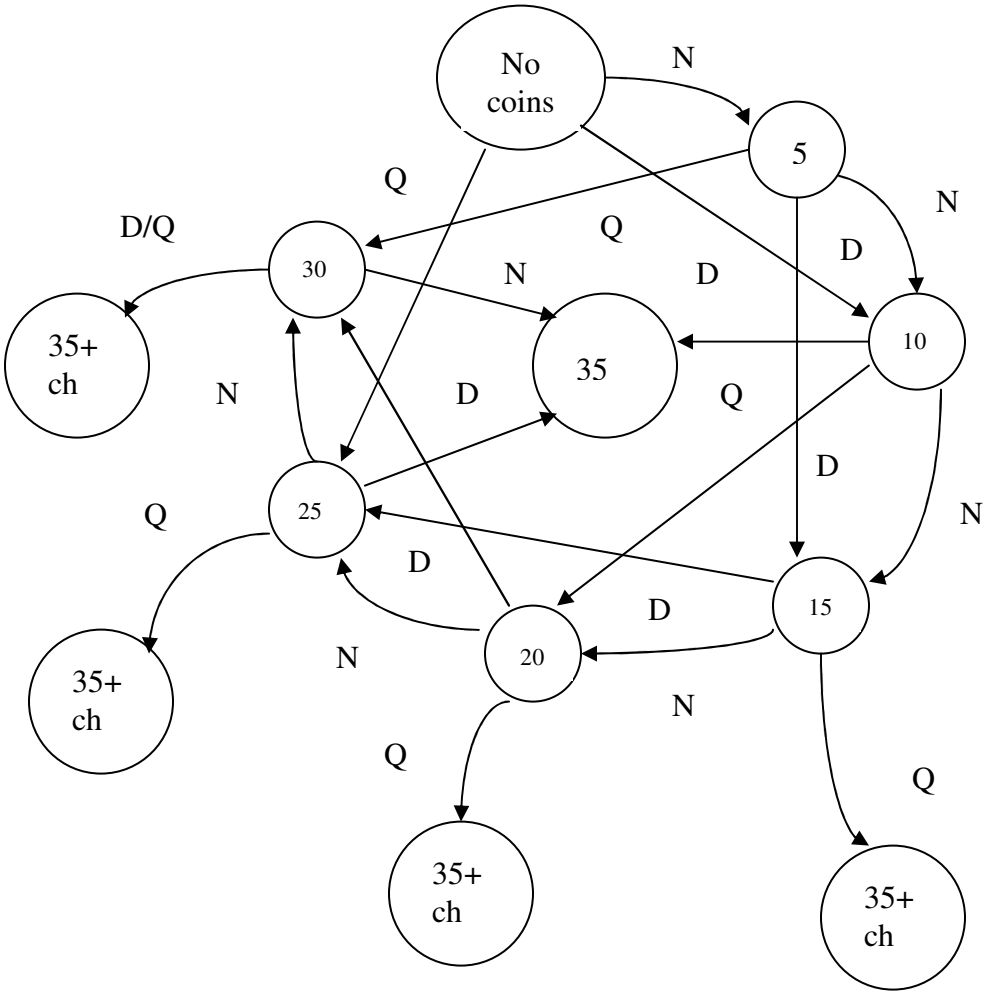
(b) Texas Win:

Texas *	Oklahoma
Fouls:10	Fouls: 10
85	70
Second Half	
0:00	
Shot Clock : 0	

(c) Oklahoma Win:

Texas *	Oklahoma
Fouls:10	Fouls: 10
81	90
First Half	
7:38	
Shot Clock : 0	

3.41


No coins
N
5
Q
N
D/Q
Q
30
D
N
D
35+ ch
10
35
N
D
Q
D
25
Q
N
D
15
D
35+ ch
N
20
N
Q
Q
35+ ch
35+ ch

3.43
a)

S1	S0	X		D1	D0	Z
0	0	0		0	0	0
0	0	1		0	0	0
0	1	0		0	0	1
0	1	1		1	0	1
1	0	0		1	1	1
1	0	1		1	1	1
1	1	0		1	0	1
1	1	1		1	0	1

b)

